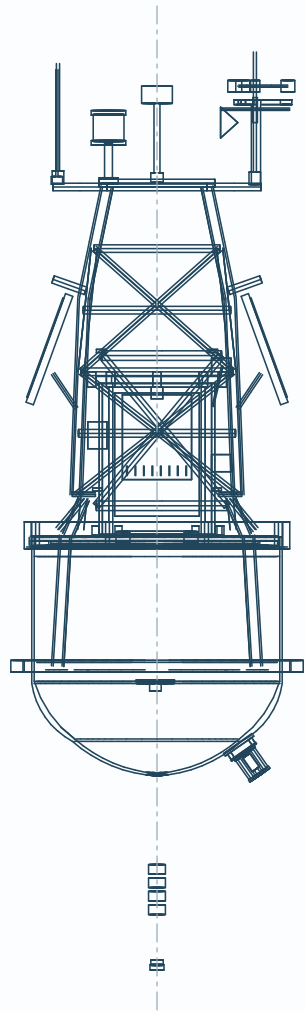


PROJECT FALCON-01 — COASTAL MONITORING BUOY

MASTER ASSEMBLY · SUBSYSTEMS · ELECTRONICS · POWER · MOORING · V2 CAD REFERENCE

REFERENCE / NOT FOR FABRICATION

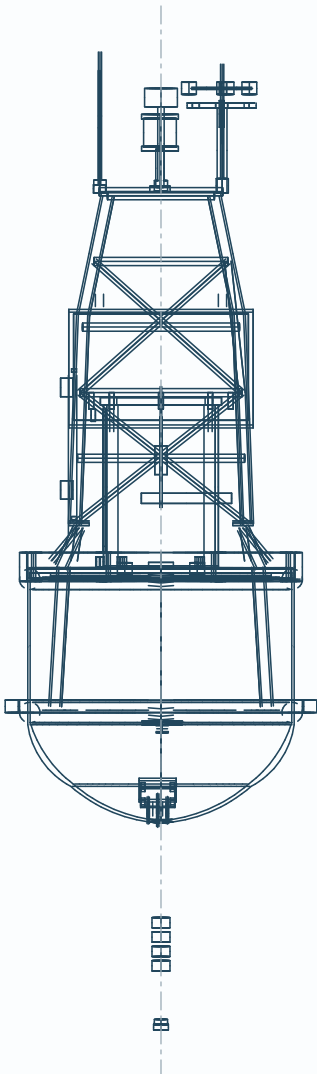
A — FRONT ELEVATION · CAD-PROJECTED



FLOAT Ø0.650 m · V2 CAD REF

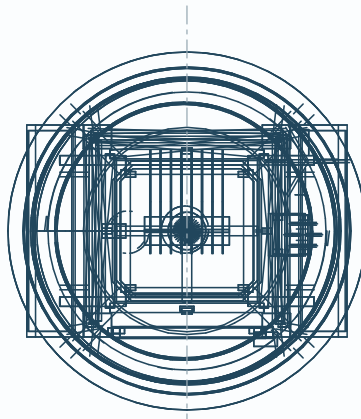
Overall installed height and loaded waterline: VERIFY IN NATIVE CAD / FLOTATION TEST

B — RIGHT ELEVATION

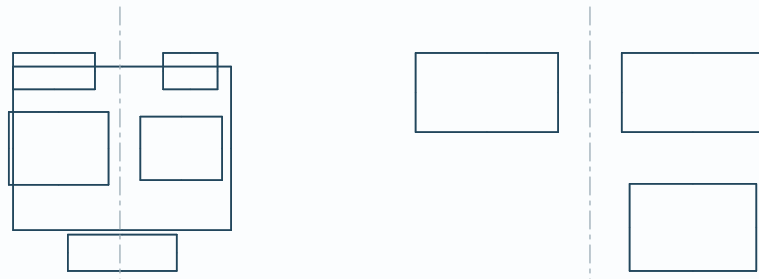


ORTHOGRAPHIC · DO NOT SCALE

C — PLAN VIEW



D — ELECTRONICS POD / EQUIPMENT PLAN



LOWER POWER DECK

UPPER CONTROL DECK

D1 BATTERY · D2 BMS · D3 MPPT

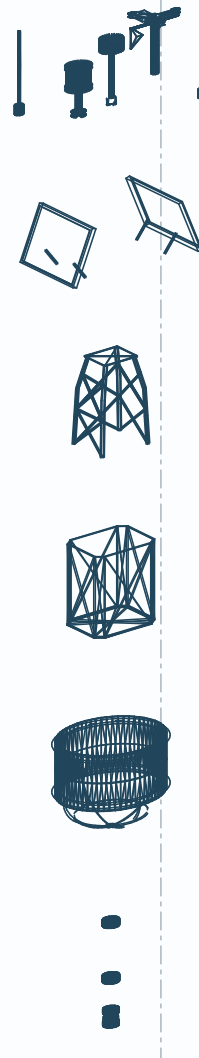
D6 ESP32 · D7 SENSOR I/O

D4 FUSE/DISCONNECT · D5 DC-DC

D8 LTE MODEM · locking headers

CAD equipment envelopes · Orange Pi excluded (shore Bay Station only)

E — EXPLODED ASSEMBLY VIEW · NTS



E01
SENSOR PLATFORM

E02
DUAL SOLAR ARRAY

E03
TAPERED FRAME

E04
SEALED ELECTRONICS POD

E05
FLOAT / KEEL

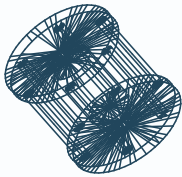
E06
BALLAST + MOORING

Exploded reference only · separation gaps are not dimensions

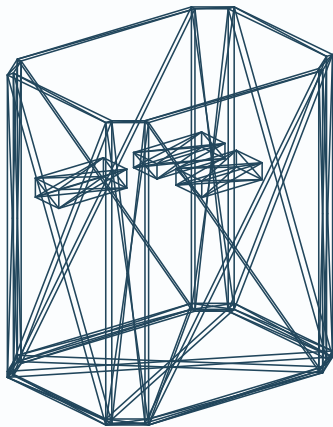
F — SUBSYSTEM INTERCONNECTION / DATA FLOW



TOP SENSOR ARRAY

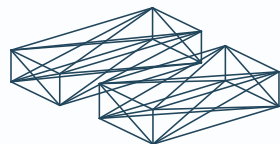


PRESSURE SENSOR ASSEMBLY

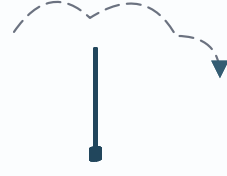


SEALED ELECTRONICS POD

protected I/O · ESP32 · cellular modem



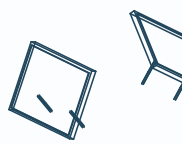
CONTROL / I/O



CELLULAR TELEMETRY

SENSORS → PROTECTED POD I/O → ESP32 ACQUISITION → LTE/INTERNET → SHORE BAY STATION / DASHBOARD

G — POWER MANAGEMENT

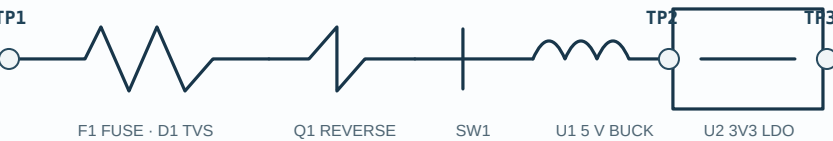


P1 · 2 × 30 W SOLAR

P2 · MPPT

P3 · LiFePO4

VBAT POWER PATH



F1 FUSE · D1 TVS

Q1 REVERSE

SW1

U1 5 V BUCK

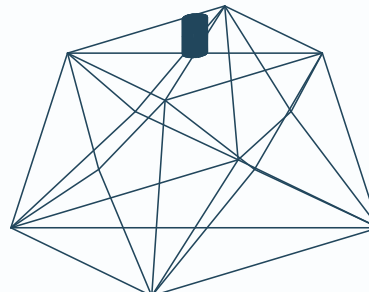
U2 3V3 LDO

TP1 VBAT · TP2 +5 V · TP3 +3V3 · final ratings controlled by measured load test

H — MOORING / ANCHORAGE



CHAIN / CONNECTING HARDWARE



CONCRETE ANCHOR · CAD REF

Scope, WLL, corrosion allowance and anchor mass remain site-engineering verification items.

J — CONTROLLED COMPONENT REGISTER

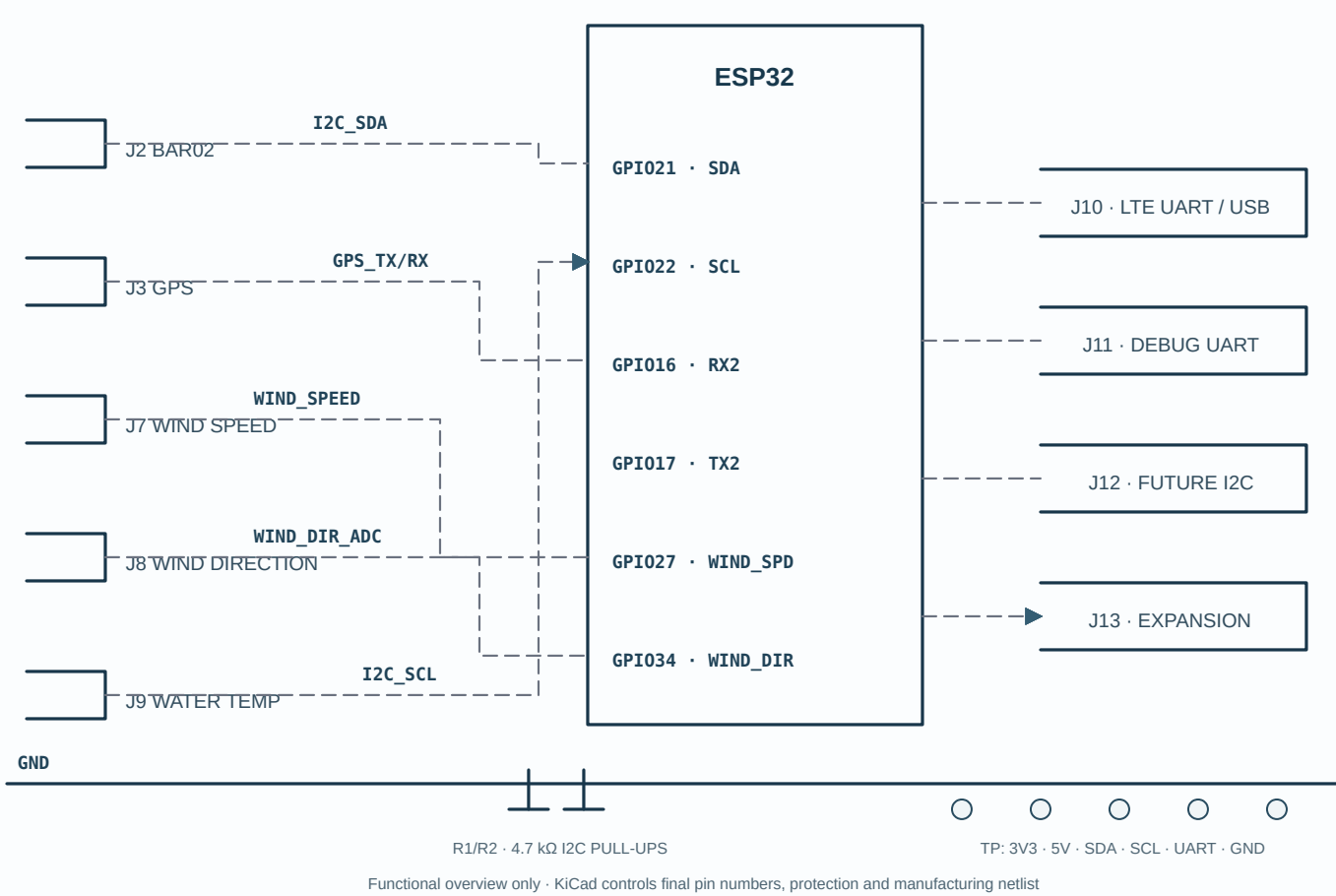
ID	COMPONENT	STATUS	FUNCTION
S01	Blue Robotics Bar02 R2	PROTOTYPE	Pressure / wave input
S02	Adafruit Ultimate GPS PID 746	PROTOTYPE	Position / geofence
S03	SparkFun SEN-15901	PROTOTYPE	Wind speed / direction
S04	Blue Robotics Celsius R2	CANDIDATE	Water temperature
H01	Adafruit MCP9808 PID 1782	SELECTED	Enclosure temperature
H02	Adafruit INA260 PID 4226 ×2	PROTOTYPE	Battery / solar health
C01	ESP32-DevKitC V4	SELECTED	Buoy controller
C02	SIM7600G-H 4G HAT	CANDIDATE	Cellular telemetry
P01	LiFePO4 12.8 V 20 Ah	RATING SET	Exact SKU / case TBD
SEC	Contact switch PID 375	PROTOTYPE	Enclosure-open state
A01	Buzzer + driver	TBD	Audible security alarm

K — SENSOR / CONTROLLER INTERFACE SCHEMATIC

FIELD CONNECTORS

U3 · ESP32-DEVKITC V4

COMMUNICATION / SERVICE



R1/R2 · 4.7 kΩ I2C PULL-UPS

TP: 3V3 · 5V · SDA · SCL · UART · GND

Functional overview only · KiCad controls final pin numbers, protection and manufacturing netlist

L — GENERAL NOTES

- Orthographic linework is projected from PROJECT-FALCON-V2.glb.
- V2 geometry remains a proposed replacement reference.
- Do not scale this drawing; verify native CAD and purchased parts.
- Dimensions, mass and placement marked TBD must not be invented.
- Complete stability, structure, ingress, thermal and mooring reviews.
- Orange Pi/Bay Station computer remains ashore.

PRELIMINARY ENGINEERING REFERENCE

M — DRAWING CONTROL

PROJECT

DRAWING TITLE

FALCON-01

MASTER ASSEMBLY & SYSTEMS

DRAWING NO.

FALCON-BP-000

REVISION

P0

SCALE

NTS

DATE

2026-09-01

SOURCE

V2 GLB + DOCS

STATUS

REFERENCE

RELEASE GATES / APPROVAL

- ☐ Native CAD dimension audit ☐ Purchased-part fit / connector clearance ☐ Loaded mass / displacement / waterline / freeboard
- ☐ CG / stability / righting test ☐ Structural load and fastener review ☐ IP / leak / thermal / corrosion validation
- ☐ Solar / battery measured load budget ☐ Pressure installation and wave-reference validation ☐ Mooring WLL / anchor design review

PREPARED BY / DATE

ENGINEERING REVIEW / DATE

THESIS ADVISER APPROVAL / DATE

NOT FOR FABRICATION